Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 2 · SLO 2 — Solved Worksheet

Name: DEEP SAND DEEP SAND | **Reg. No:** RA2511026010557

Unit â€“ 1 : Session â€“ 2 : SLO â€“ 2 | Array of Structures

I. Fill in the Blanks

1. multiple records â€” an array of structures stores many records of the same type.
2. arrayName[size] â€” struct StructName arr[100];
3. arr[index].member â€” access a member via index and dot operator.
4. 5 Student structures â€” struct Student s[5]; declares an array of 5 Student records.
5. student/employee records â€— e.g. roll numbers, names, salaries.
6. loop (for loop) â€— read values into the array with a loop and scanf.
7. emp[2].name â€— the 3rd element's name field (index 2).
8. normal structure variable â€— each element behaves like a regular struct variable.
9. arr (or &arr[0]) â€— struct StructName *ptr = arr;
10. for loop â€— loop with the dot operator prints each record's details.

II. Code Snippets

1. 100 students: struct Student { int roll; char name[50]; }; struct Student s[100]; â€— loop i=0..99 with scanf("%d %s", &s[i].roll, s[i].name).
2. 5 employees: struct Emp { int id; float salary; }; struct Emp e[5]; â€— input loop then display loop printing e[i].id, e[i].salary.
3. 10 students Ã— 3 subjects: struct Result { int marks[3]; }; struct Result r[10]; â€— r[i].marks[j] holds subject j of student i.
4. Shopping products: struct Product { char name[50]; float price; }; struct Product p[50];
5. Cricket team: struct Player { char name[50]; int runs; }; struct Player team[11];

III. Practice the Code

• struct Car { char model[30]; int year; };
• struct Car c[3] = {{"Swift", 2020}, {"City", 2022}, {"Creta", 2023}};
• for (int i = 0; i < 3; i++) printf("%s (%d)\n", c[i].model, c[i].year);